

# SynDiff Synthetic Benchmark — Live Progress

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**942/942** jobs complete (100.0%)

$S = I(X_1; Y | X_2) - I(X_1; Y)$  in nats;  $S > 0$  synergy-dominant,  $S < 0$  redundancy-dominant. SynDiff (ours) vs prior estimators.

## Dataset construction

### Phase 1 — Gaussian (closed-form ground truth)

Scalar base:  $X_1, X_2 \sim \mathcal{N}(0, 1)$  with  $\text{corr}(X_1, X_2) = \rho$ ; label  $Y = X_1 + X_2 + \sigma_y \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, 1)$ . Thus  $\text{Var}(Y) = 2 + 2\rho + \sigma_y^2$  and  $\text{Cov}(X_i, Y) = 1 + \rho$ . To lift to  $d$  dimensions the informative scalar is padded with  $d - 1$  i.i.d.  $\mathcal{N}(0, 1)$  coordinates and rotated by a fixed per-seed orthogonal matrix  $Q_i$  (one per variable). Rotation and independent padding preserve mutual information, so  $S$  is unchanged and the closed form carries over. Ground truth is computed from the  $3 \times 3$  covariance via  $I = \frac{1}{2} \log(\det \Sigma_A \det \Sigma_B / \det \Sigma_{AB})$  and the analogous conditional form. Cells sweep  $\rho \in \{-.9, \dots, .9\}$ ,  $\sigma_y \in \{.1, 1\}$ ,  $n \in \{1\text{k}, \dots, 100\text{k}\}$ ,  $d \in \{1, 8, 32, 128\}$ .

### Phase 2 — discrete U/R/S (mixture-density MC truth)

Latent sources  $s_1, s_2, s_3 \sim \text{Unif}\{0, \dots, Q - 1\}$  with  $Q = 8$ . Encodings  $X_i = W_i \text{onehot}(s_i) + \sigma_{\text{enc}} \mathcal{N}(0, I_{32})$  with orthonormal  $W_i \in \mathbb{R}^{32 \times 8}$  frozen per seed. Regimes ( $X_1$  always encodes  $s_1$ ): **UNIQUE**  $Y = s_1$ ; **REDUNDANT**  $Y = s_1$  and  $X_2$  also encodes  $s_1$ ; **SYNERGY1**  $Y = (s_1 + s_2) \bmod Q$  with  $X_2 \sim s_2$ ; **SYNERGY2**  $Y = (s_3 + (s_1 + s_2) \bmod Q) \bmod Q$  with  $X_2 = [\text{enc}(s_2) | \text{enc}(s_3)] \in \mathbb{R}^{64}$ ; **INDEPENDENT**  $Y = s_3$ . The  $\lambda$ -dial mixes the SYNERGY1 and REDUNDANT rules per sample (Bernoulli( $\lambda$ )). Ground truth is the exact  $S = I(X_1; X_2, Y) - I(X_1; X_2) - I(X_1; Y)$  obtained by Monte Carlo over the enumerable Gaussian-mixture components (matches the discrete limit as  $\sigma_{\text{enc}} \rightarrow 0$ :  $\pm \log Q$  for pure synergy/redundancy, 0 for unique/independent).

## Method: SynDiff

Target  $S = I(X_1; Y | X_2) - I(X_1; Y)$  in nats. By the symmetry of interaction information this equals  $I(X_1; X_2 | Y) - I(X_1; X_2)$ , which the four-mask form below estimates directly. **VP schedule:**  $\beta(t) = 0.1 + 19.9t$ ,  $B(t) = \int_0^t \beta = 0.1t + 9.95t^2$ , signal  $k(t) = e^{-B/2}$ , noise  $\sigma^2(t) = 1 - e^{-B}$ , weight  $w(t) = \beta(t)/(2\sigma^2(t))$ . A single noise-prediction network  $\hat{e}(x_{1,t}, t, \text{cond})$  (*MaskedCondMLP*: Fourier- $t$  features  $\oplus$  per-slot conditioning embeddings each with a learned NULL token;  $4 \times 512$  SiLU) is trained to denoise  $X_1$  while each conditioning slot ( $X_2, Y$ ) is independently dropped to its NULL token with probability 0.15. EMA weights (decay 0.999) are used at evaluation. **Four-mask estimator.** Per evaluation triple and per stratified time  $t$  ( $K=16$  strata on  $[10^{-3}, 1]$ ) with *one shared* noise draw  $\varepsilon$  and  $x_{1,t} = k(t)x_1 + \sigma(t)\varepsilon$ :

$$\begin{aligned} e_1 &= \hat{e}(x_{1,t}, t, x_2, y), & e_2 &= \hat{e}(x_{1,t}, t, \emptyset, y), \\ e_3 &= \hat{e}(x_{1,t}, t, x_2, \emptyset), & e_4 &= \hat{e}(x_{1,t}, t, \emptyset, \emptyset), \end{aligned}$$

$$\text{integrand}(t) = w(t)(\|e_1 - e_2\|^2 - \|e_3 - e_4\|^2).$$

Per-sample  $\hat{S} = (1 - 10^{-3}) \overline{\text{integrand}_t}$ ; the estimate is the mean over  $10^4$  fresh triples with a 1000-sample bootstrap CI (norms in fp32, no-grad, EMA weights). **Baselines:** *minde\_comp* (three

separately trained MINDE MI terms composed as above), *MINE* and *InfoNCE* (lower-bound MI, same composition), *batch\_pid* (Liang et al. CVX PID, k-means  $k=20$  then max-entropy solve), and *broja/plugin* on  $d=1$ .

## Sweep progress

| Phase | Method     | Done | Expected |
|-------|------------|------|----------|
| P1    | batch_pid  | 96   | 96       |
| P1    | broja      | 6    | 6        |
| P1    | infonce    | 96   | 96       |
| P1    | minde_comp | 96   | 96       |
| P1    | mine       | 96   | 96       |
| P1    | plugin     | 6    | 6        |
| P1    | syndiff    | 96   | 96       |
| P2    | batch_pid  | 90   | 90       |
| P2    | infonce    | 90   | 90       |
| P2    | minde_comp | 90   | 90       |
| P2    | mine       | 90   | 90       |
| P2    | syndiff    | 90   | 90       |

## Accuracy vs ground truth (completed cells, lower error better)

| Method         | #cells | mean $ \text{est} - \text{truth} $ | sign acc |
|----------------|--------|------------------------------------|----------|
| broja          | 2      | 0.0680                             | 1.00     |
| plugin         | 2      | 0.1301                             | 1.00     |
| minde_comp     | 62     | 0.1419                             | 0.81     |
| <b>syndiff</b> | 62     | <b>0.2277</b>                      | 0.89     |
| batch_pid      | 62     | 0.3660                             | 0.74     |
| mine           | 62     | 0.8132                             | 0.58     |
| infonce        | 62     | 3.9327                             | 0.37     |

## SynDiff vs prior — absolute error to truth $|\hat{S} - S|$ (nats)

Each entry is  $|\text{estimate} - \text{truth}|$  in nats; **bold** marks the most accurate method for that cell. ‘—’ means not yet computed.

| Cell                         | truth  | <b>SynDiff</b> | batch_pid    | minde_comp   | mine  | infonce | plugin |
|------------------------------|--------|----------------|--------------|--------------|-------|---------|--------|
| p1_dial_rho0_sy0p1_d8_n20k   | +1.963 | 0.076          | 1.817        | <b>0.041</b> | 1.332 | 7.567   | —      |
| p1_dial_rho0_sy1_d8_n20k     | +0.144 | 0.008          | 0.007        | <b>0.001</b> | 0.699 | 5.459   | —      |
| p1_dial_rho0p3_sy0p1_d8_n20k | +1.740 | 0.063          | 1.609        | <b>0.062</b> | 1.113 | 7.200   | —      |
| p1_dial_rho0p3_sy1_d8_n20k   | +0.007 | <b>0.001</b>   | 0.121        | 0.003        | 0.610 | 5.436   | —      |
| p1_dial_rho0p5_sy0p1_d8_n20k | +1.477 | 0.071          | 1.367        | <b>0.043</b> | 0.952 | 6.914   | —      |
| p1_dial_rho0p5_sy1_d8_n20k   | -0.134 | <b>0.001</b>   | 0.244        | 0.005        | 0.695 | 5.120   | —      |
| p1_dial_rho0p7_sy0p1_d8_n20k | +1.035 | 0.064          | 0.955        | <b>0.045</b> | 0.726 | 6.430   | —      |
| p1_dial_rho0p7_sy1_d8_n20k   | -0.329 | <b>0.005</b>   | 0.417        | 0.008        | 0.322 | 4.861   | —      |
| p1_dial_rho0p9_sy0p1_d8_n20k | +0.024 | 0.057          | <b>0.024</b> | 0.050        | 0.353 | 5.345   | —      |
| p1_dial_rho0p9_sy1_d8_n20k   | -0.610 | 0.010          | 0.672        | <b>0.002</b> | 0.196 | 4.535   | —      |

| Cell                          | truth  | SynDiff      | batch_pid    | minde_comp   | mine         | infonce      | plugin |
|-------------------------------|--------|--------------|--------------|--------------|--------------|--------------|--------|
| p1_dial_rhon0p3_sy0p1_d8_n20k | +2.047 | 0.071        | 1.891        | <b>0.031</b> | 1.323        | 7.433        | –      |
| p1_dial_rhon0p3_sy1_d8_n20k   | +0.209 | 0.010        | 0.054        | <b>0.001</b> | 0.771        | 5.572        | –      |
| p1_dial_rhon0p5_sy0p1_d8_n20k | +2.023 | 0.070        | 1.872        | <b>0.043</b> | 1.228        | 7.530        | –      |
| p1_dial_rhon0p5_sy1_d8_n20k   | +0.213 | 0.016        | 0.061        | <b>0.001</b> | 0.717        | 5.555        | –      |
| p1_dial_rhon0p7_sy0p1_d8_n20k | +1.896 | 0.070        | 1.760        | <b>0.042</b> | 1.207        | 7.384        | –      |
| p1_dial_rhon0p7_sy1_d8_n20k   | +0.177 | 0.019        | 0.029        | <b>0.005</b> | 0.825        | 5.552        | –      |
| p1_dial_rhon0p9_sy0p1_d8_n20k | +1.473 | 0.062        | 1.334        | <b>0.028</b> | 1.008        | 6.944        | –      |
| p1_dial_rhon0p9_sy1_d8_n20k   | +0.083 | 0.029        | 0.060        | <b>0.006</b> | 0.663        | 5.283        | –      |
| p1_dim_rho0p7_sy1_d128_n20k   | -0.329 | 0.278        | 0.469        | <b>0.034</b> | 1.524        | 5.804        | –      |
| p1_dim_rho0p7_sy1_d1_n20k     | -0.329 | <b>0.000</b> | 0.079        | 0.006        | 0.008        | 0.058        | 0.137  |
| p1_dim_rho0p7_sy1_d32_n20k    | -0.329 | 0.036        | 0.463        | <b>0.002</b> | 1.138        | 5.773        | –      |
| p1_dim_rho0p7_sy1_d128_n20k   | +0.177 | 0.155        | <b>0.039</b> | 0.094        | 2.115        | 6.320        | –      |
| p1_dim_rho0p7_sy1_d1_n20k     | +0.177 | 0.009        | 0.055        | 0.007        | 0.006        | <b>0.006</b> | 0.123  |
| p1_dim_rho0p7_sy1_d32_n20k    | +0.177 | 0.048        | 0.031        | <b>0.000</b> | 1.262        | 6.283        | –      |
| p1_ns_rho0p7_sy1_d8_n100k     | -0.329 | <b>0.003</b> | 0.417        | 0.007        | 0.375        | 4.876        | –      |
| p1_ns_rho0p7_sy1_d8_n1k       | -0.329 | <b>0.002</b> | 0.417        | 0.004        | 0.301        | 4.888        | –      |
| p1_ns_rho0p7_sy1_d8_n20k      | -0.329 | <b>0.004</b> | 0.417        | 0.006        | 0.323        | 4.932        | –      |
| p1_ns_rho0p7_sy1_d8_n5k       | -0.329 | <b>0.003</b> | 0.417        | 0.008        | 0.322        | 4.846        | –      |
| p1_ns_rho0p7_sy1_d8_n100k     | +0.177 | 0.020        | 0.028        | <b>0.003</b> | 0.825        | 5.552        | –      |
| p1_ns_rho0p7_sy1_d8_n1k       | +0.177 | 0.017        | 0.029        | <b>0.006</b> | 0.674        | 5.515        | –      |
| p1_ns_rho0p7_sy1_d8_n20k      | +0.177 | 0.023        | 0.029        | <b>0.003</b> | 0.779        | 5.552        | –      |
| p1_ns_rho0p7_sy1_d8_n5k       | +0.177 | 0.023        | 0.029        | <b>0.008</b> | 0.630        | 5.626        | –      |
| p2_independent_senc0p1        | +0.000 | <b>0.000</b> | 0.142        | 0.001        | 0.305        | 1.923        | –      |
| p2_independent_senc0p5        | +0.000 | <b>0.000</b> | 0.141        | 0.000        | 0.763        | 2.083        | –      |
| p2_independent_senc1          | +0.000 | <b>0.000</b> | 0.138        | 0.000        | 0.844        | 2.070        | –      |
| p2_lambda0_senc0p1            | -2.079 | 4.088        | 0.025        | 0.511        | <b>0.016</b> | 0.112        | –      |
| p2_lambda0_senc0p5            | -0.730 | 0.243        | 0.521        | <b>0.084</b> | 0.279        | 1.292        | –      |
| p2_lambda0_senc1              | -0.065 | 0.065        | 0.210        | <b>0.017</b> | 0.909        | 2.019        | –      |
| p2_lambda0p25_senc0p1         | -0.302 | 0.136        | <b>0.025</b> | 0.504        | 0.407        | 1.979        | –      |
| p2_lambda0p25_senc0p5         | -0.134 | <b>0.004</b> | 0.160        | 0.090        | 0.626        | 1.928        | –      |
| p2_lambda0p25_senc1           | -0.016 | 0.017        | 0.159        | <b>0.002</b> | 0.764        | 2.021        | –      |
| p2_lambda0p5_senc0p1          | +0.814 | 0.194        | <b>0.019</b> | 0.534        | 0.676        | 2.956        | –      |
| p2_lambda0p5_senc0p5          | +0.272 | 0.203        | <b>0.040</b> | 0.170        | 0.826        | 2.347        | –      |
| p2_lambda0p5_senc1            | +0.024 | <b>0.020</b> | 0.120        | 0.020        | 0.977        | 2.103        | –      |
| p2_lambda0p75_senc0p1         | +1.583 | 0.137        | <b>0.027</b> | 0.479        | 0.953        | 3.646        | –      |
| p2_lambda0p75_senc0p5         | +0.545 | 0.212        | 0.169        | <b>0.154</b> | 1.363        | 2.635        | –      |
| p2_lambda0p75_senc1           | +0.048 | 0.044        | 0.095        | <b>0.037</b> | 0.572        | 2.106        | –      |
| p2_lambda1_senc0p1            | +2.079 | 0.204        | <b>0.013</b> | 0.747        | 1.249        | 4.239        | –      |
| p2_lambda1_senc0p5            | +0.733 | 0.173        | 0.256        | <b>0.125</b> | 1.553        | 2.749        | –      |
| p2_lambda1_senc1              | +0.067 | <b>0.067</b> | 0.078        | 0.067        | 0.862        | 2.167        | –      |
| p2_redundant_senc0p1          | -2.079 | 3.678        | 0.027        | 1.084        | <b>0.021</b> | 0.115        | –      |
| p2_redundant_senc0p5          | -0.730 | 0.305        | 0.523        | <b>0.096</b> | 0.194        | 1.305        | –      |
| p2_redundant_senc1            | -0.065 | 0.065        | 0.199        | <b>0.024</b> | 0.703        | 2.006        | –      |
| p2_synergy1_senc0p1           | +2.079 | 0.172        | <b>0.013</b> | 0.726        | 1.255        | 4.113        | –      |
| p2_synergy1_senc0p5           | +0.733 | 0.187        | 0.251        | <b>0.124</b> | 1.651        | 2.827        | –      |
| p2_synergy1_senc1             | +0.067 | <b>0.067</b> | 0.078        | 0.067        | 1.081        | 2.121        | –      |
| p2_synergy2_senc0p1           | +2.079 | 2.079        | <b>1.362</b> | 2.081        | 1.457        | 4.048        | –      |
| p2_synergy2_senc0p5           | +0.424 | 0.424        | <b>0.272</b> | 0.424        | 1.294        | 2.508        | –      |
| p2_synergy2_senc1             | +0.010 | 0.010        | 0.132        | <b>0.010</b> | 0.758        | 2.074        | –      |
| p2_unique_senc0p1             | -0.000 | 0.000        | <b>0.000</b> | 0.009        | 0.838        | 2.042        | –      |
| p2_unique_senc0p5             | -0.000 | <b>0.000</b> | 0.142        | 0.005        | 1.128        | 2.075        | –      |
| p2_unique_senc1               | -0.000 | <b>0.000</b> | 0.142        | 0.002        | 1.070        | 2.071        | –      |
| # cells most accurate         |        | <b>17</b>    | 11           | 31           | 2            | 1            | 0      |